

Sensor Dropout on CH_31B Resolved — Conveyor Assembly, Loose Terminal at J-301

Asset: AST_CON_001 (Conveyor) | Bearing: BRG_009 (SKF22212)
Fault Mode: sensor_fault | ISO Stage: No Stage (Healthy/Sensor)
Learned Case — Knowledge Agent Retrieval Document | DRO Baseline
CASE_003 | Outcome: SUCCESSFUL | Opened: 2026-05-02 | Closed: 2026-05-02

1. Detection Summary

Detection Date	2026-05-02
Detected By	data_foundation_agent (signal_quality_event)
Anomaly Score	n/a (data quality event, not a bearing fault)
Dropout Pattern	Burst — 2 to 4 consecutive null readings, then recovery
Dropout Percentage	3.2% of readings over 7-day monitoring window
signal_quality_score during dropout	0.10 to 0.15 (healthy: ≥ 0.97)
Channel Affected	CH_31B (historian tag: PI:CON001.BRG9:VIB)
BRG_010 (NON_DRIVE)	Clean signal — dropout was channel-specific, not asset-wide
LOTO Status at Detection	CRITICAL: LOTO_003 expired 2026-04-15 — work held pending renewal

2. Agent Diagnosis

Primary Diagnosis	signal_fault — loose connector (intermittent burst pattern)
Bearing Diagnosis Confidence	Reduced to 71% (from typical 85%) — unreliable channel data
Reasoning	Burst-pattern dropouts are characteristic of a loose terminal or damaged cable connector. Random single-record dropouts indicate transient noise. Burst pattern = intermittent physical connection.
Bearing Condition	Could not be confirmed while CH_31B was unreliable

3. Action Taken

Recommended Action	Sensor inspection and cable repair at junction box J-301; visual bearing inspection while accessed
LOTO Constraint	New LOTO_003 permit required before work could start — caused 1-day delay
Window Selected	LINE_001 planned stop, same-day after permit renewal
Parts Required	Replacement cable connector from OT instrument store
Crew Assigned	Raj Kumar (TECH_006, OT instrument specialist)
Work Order	WO_003 — priority HIGH (data integrity risk on bottleneck asset)

4. Findings at Inspection / Teardown

Junction box J-301 (non-drive end, conveyor frame) was opened. The cable termination for the CH_31B signal conductor was found with the terminal screw partially backed out — approximately half a turn loose. The terminal had been vibrating loose over time due to conveyor operating vibration. No cable damage was found. The terminal was re-terminated and tightened to 6 Nm with a calibrated torque screwdriver. Visual inspection of BRG_009 (DRIVE bearing) confirmed normal condition — no visible damage, grease leakage, or discolouration, consistent with

health_score 0.72. BRG_010 (NON_DRIVE) also appeared normal.

5. Outcome

Signal Quality Post-Repair	0.98 (15 consecutive clean readings confirmed)
Null Records in 4h Post-Repair	Zero
BRG_009 Vibration (channel restored)	2.3 mm/s (within baseline)
QA Result	PASS — signal quality restored, no bearing fault found
Actual Duration	3 hours (including permit renewal wait time)
Actual Cost	INR 3,000
Recommendation Followed	Yes

6. Root Cause Confirmed

Root Cause	loose_connector_J301
Contributing Factors	Terminal screw not torqued to specification during original sensor installation. Conveyor vibration gradually backed the screw out over approximately 6 months.

7. Lessons Learned

- Burst-pattern signal dropouts (consecutive nulls in clusters) reliably indicate a physical connection problem — not a bearing fault. Data Foundation Agent correctly classified this.
- Expired LOTO permits are a hard stop. The 1-day delay caused by LOTO_003 renewal is operationally significant on a bottleneck asset — permits should be renewed proactively before expiry.
- Terminal tightening torque (5–6 Nm) must be verified at every sensor installation using a calibrated torque screwdriver — not by feel.
- CH_31B (historian tag: PI:CON001.BRG9:VIB) must not be renamed — this ID is fixed in the telemetry schema. BRG_010 on CH_010 remained clean, confirming the dropout was channel-specific.